A FILTRATION BOUND FOR THE ERASURE ENTROPIES OF BOOLEAN FUNCTIONS

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Abstract. For a Boolean function f and a coordinate i, the erasure entropy $E_i(f)$ is the conditional entropy of the event $i \in U$ given $U \setminus \{i\}$, where U is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture is equivalent to the existence of an absolute c such that every non-constant f has a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$. We bound E_i from above by revealing the other coordinates one at a time and conditioning only on the last revealed coordinate that the spectral sample contains. The resulting quantity L_i^π depends on the order π, is a sum, over the steps, of a two-variable entropy of the squared increments of the Doob martingales of $E_i f$ and $D_i f$, lies between E_i and the binary entropy of Inf_i, and can be minimised over all orders by a shortest-path recursion. We show that $L_i^\pi = O(\mathrm{Inf}_i)$ whenever, along the order, the probability that i is pivotal both before and after the not yet revealed coordinates are resampled, times the block influence of those coordinates, is $O(\mathrm{Inf}_i^2)$. We also give an exact formula for E_i under block composition, and show that a coordinate whose variable enters only through one AND of $k+1$ variables, pivotal with probability ρ, has $L_i^\pi/\mathrm{Inf}_i \le 2 + 2^{-k}(2k + \log_2(1/\rho))$ for a suitable order. Conditioning on the last r revealed coordinates gives a decreasing family of bounds that reaches E_i at $r = n-1$.

1. Introduction

For $f\colon \{-1,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$, and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [1] asks for an absolute C with $H(f) \le C\,I(f)$ for every Boolean f. It is known for symmetric functions and read-once decision trees [8], it is preserved under suitable compositions [7], and there are bounds in terms of decision-tree and communication complexity measures [3, 13]; Arunachalam, Chakraborty, Koucký, Saurabh and de Wolf [2] bound $H(f)$ by twice the average unambiguous parity-certificate complexity. Kelman, Kindler, Lifshitz, Minzer and Safra [6] prove a weak form for the min-entropy, and the best general bound is Han's [5]; a form of it with $\mathrm{Var}(f)/\mathrm{Inf}_i(f)$ in place of $1/\mathrm{Inf}_i(f)$ in the logarithm is in [11].

One route to the conjecture goes through a single coordinate at a time. The drop of the entropy from f to its two restrictions at $x_i = \pm 1$ splits into the erasure entropy (in the sense of Verdú and Weissman [12])

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big)$$

and a Jensen–Shannon term $J_i(f) \le (2/\ln 2)\mathrm{Inf}_i(f)$. The route goes back to Han [5, Remarks 3.1–3.2]; the splitting is equation (32) of González, MacManus and Pereyra [4]. In [9] it is shown that the conjecture holds if and only if there is an absolute c such that every non-constant Boolean function has a coordinate i with

$$E_i(f) \le c\,\mathrm{Inf}_i(f), \tag{1}$$

and in [10] this quantity is studied directly. The difficulty is that E_i conditions on the whole of $U \setminus \{i\}$, an object with up to 2^{n-1} values. The trivial bound $E_i \le H(\mathbf{1}[i \in U])$ loses a factor $\log(1/\mathrm{Inf}_i)$, and (1) asks exactly to remove that factor at one coordinate.

Here we condition on less. Reveal the other coordinates in a fixed order and record only the position of the last revealed coordinate that U contains. This gives an upper bound L_i^π for E_i (Theorem 1) with a martingale form: at each step it compares how much of the restriction $E_i f$ and how much of the derivative $D_i f$ the newly revealed coordinate explains. A two-sequence inequality turns a product condition along the order into the bound $L_i^\pi = O(\mathrm{Inf}_i)$ (Theorem 4), and the product condition has a direct reading in terms of resampling. Section 4 gives an exact composition formula for E_i, Section 5 the case of a private AND, and Section 6 the windows of length r. Section 7 lists exact values.

2. The witness

Fix i and write $g = E_i f$ and $h = D_i f$ for the functions of the other coordinates

$$g = \tfrac{1}{2}\big(f|_{x_i=1} + f|_{x_i=-1}\big), \qquad h = \tfrac{1}{2}\big(f|_{x_i=1} - f|_{x_i=-1}\big).$$

They take values in $\{-1, 0, 1\}$, have disjoint supports, $\hat{g}(T) = \hat{f}(T)$ and $\hat{h}(T) = \hat{f}(T \cup \{i\})$ for $T \not\ni i$, $\mathbb{E}h^2 = \mathrm{Inf}_i(f) =: \delta$ and $\mathbb{E}g^2 = 1 - \delta$. Put $p(T) = \hat{f}(T)^2$ and $q(T) = \hat{f}(T \cup \{i\})^2$ for $T \subseteq [n] \setminus \{i\}$, and

$$\Phi(a, b) = \eta(a) + \eta(b) - \eta(a + b) = (a + b)\, h_2\Big(\frac{b}{a + b}\Big), \qquad \eta(x) = -x \log_2 x,$$

where h_2 is the binary entropy. Then $E_i(f) = \sum_T \Phi(p(T), q(T))$, since $(U \setminus \{i\}, \mathbf{1}[i \in U])$ takes the value $(T, 0)$ with probability $p(T)$ and $(T, 1)$ with probability $q(T)$. The function Φ is concave, homogeneous of degree one and increasing in each argument; in particular it is superadditive.

Let $\pi = (j_1, \dots, j_m)$, $m = n-1$, be an order of the other coordinates, and let $S_k = \{j_1, \dots, j_k\}$. For $T \subseteq [n] \setminus \{i\}$ let $\mathrm{lab}(T) = \max\{k : j_k \in T\}$, with $\mathrm{lab}(\emptyset) = 0$, and put

$$A_k = \sum_{\mathrm{lab}(T)=k} p(T), \qquad B_k = \sum_{\mathrm{lab}(T)=k} q(T), \qquad L_i^\pi(f) = \sum_{k=0}^{m} \Phi(A_k, B_k).$$

Theorem 1. *For every Boolean f, coordinate i and order π:*

(a) $E_i(f) \le L_i^\pi(f) \le h_2(\mathrm{Inf}_i(f))$.

(b) *With $\mathcal{F}_k$ the σ-algebra generated by $x_{j_1}, \dots, x_{j_k}$,*

$$A_k = \mathbb{E}\big(\mathbb{E}[g \mid \mathcal{F}_k] - \mathbb{E}[g \mid \mathcal{F}_{k-1}]\big)^2, \qquad B_k = \mathbb{E}\big(\mathbb{E}[h \mid \mathcal{F}_k] - \mathbb{E}[h \mid \mathcal{F}_{k-1}]\big)^2$$

for $k \ge 1$, and $A_0 = (\mathbb{E}g)^2$, $B_0 = (\mathbb{E}h)^2$.

(c) *Let $a(R) = \sum_{T \subseteq R} p(T)$ and $b(R) = \sum_{T \subseteq R} q(T)$. Then $\min_\pi L_i^\pi = \beta(R_*)$ for the full set $R_* = [n] \setminus \{i\}$, where $\beta(\emptyset) = \Phi(a(\emptyset), b(\emptyset))$ and*

$$\beta(R) = \min_{j \in R}\Big(\beta(R \setminus \{j\}) + \Phi\big(a(R) - a(R \setminus \{j\}),\, b(R) - b(R \setminus \{j\})\big)\Big).$$

Proof. (a) The pair $(\mathrm{lab}(U \setminus \{i\}), \mathbf{1}[i \in U])$ takes the value $(k, 0)$ with probability A_k and $(k, 1)$ with probability B_k, so $L_i^\pi = H(\mathbf{1}[i \in U] \mid \mathrm{lab}(U \setminus \{i\}))$. Since $\mathrm{lab}(U \setminus \{i\})$ is a function of $U \setminus \{i\}$, conditioning on it cannot give a smaller entropy than conditioning on $U \setminus \{i\}$, and it cannot give a larger one than conditioning on nothing, which is $h_2(\delta)$.

(b) $\mathbb{E}[g \mid \mathcal{F}_k] = \sum_{T \subseteq S_k} \hat{g}(T)\chi_T$, so the k-th increment is $\sum_{\mathrm{lab}(T)=k} \hat{g}(T)\chi_T$, whose squared norm is A_k by Parseval. The same holds for h.

(c) The sets $\{T : \mathrm{lab}(T) \le k\}$ are the subsets of S_k, so $A_k = a(S_k) - a(S_{k-1})$ and $B_k = b(S_k) - b(S_{k-1})$. An order is a maximal chain $\emptyset = S_0 \subset S_1 \subset \cdots \subset S_m = R_*$, and L_i^π is a sum of costs along its edges. The recursion is the shortest path in this graph. □

The recursion costs $O(m2^m)$ operations per coordinate, against $m!$ orders. Part (b) says that L_i^π follows the Doob martingales of g and of h along the order and charges, at each step, the entropy of telling the two increments apart. A coordinate is good along π when h is explained late, after most of g has been explained.

3. A sufficient condition

Lemma 2. *Let $A_k, B_k \ge 0$ for $k = 0, \dots, m$, let $\delta = \sum_k B_k > 0$, and suppose that for some $M \ge 0$*

$$\Big(\sum_{k\ge K} A_k\Big)\Big(\sum_{k\le K} B_k\Big) \le M\delta^2 \qquad \text{for every } K = 0, \dots, m.$$

Then $\sum_k \Phi(A_k, B_k) \le \big(1/\ln 2 + 4 + 2\log_2(1+2M)\big)\delta$.

Proof. Since $a\log_2(1+b/a) \le b/\ln 2$, we have $\Phi(a,b) \le b/\ln 2 + \psi(a,b)$ with $\psi(a,b) = b\log_2(1+a/b)$. The function ψ is concave, homogeneous of degree one and increasing in each argument, so $\sum_{k\in G}\psi(A_k,B_k) \le \psi(A_G,B_G)$ for every set G of indices, where $A_G = \sum_{k\in G} A_k$. Put $\beta_k = \delta^{-1}\sum_{l\le k} B_l$, which is nondecreasing, and $G_j = \{k : 2^{-j-1} < \beta_k \le 2^{-j}\}$ for $j \ge 0$. Indices with $\beta_k = 0$ have $B_k = 0$ and contribute nothing. For $k_j = \min G_j$ the hypothesis gives $A_{G_j} \le \sum_{k\ge k_j} A_k \le M\delta/\beta_{k_j} \le 2^{j+1}M\delta$, and $B_{G_j} \le \beta_{\max G_j}\delta \le 2^{-j}\delta$. Hence

$$\sum_{k\in G_j}\psi(A_k,B_k) \le 2^{-j}\delta\log_2\big(1+2^{2j+1}M\big) \le 2^{-j}\delta\big(2j+\log_2(1+2M)\big),$$

and summing over j with $\sum_j 2^{-j} = 2$ and $\sum_j j2^{-j} = 2$ gives the claim. □

Remark 3. Under the hypothesis of Lemma 2 one also has $\sum_k \Phi(A_k,B_k) \le \big(4/\ln 2 + \log_2(1+2M)\big)\delta$, which is the smaller bound when $M \ge 0.13$. Work in natural logarithms, put $c = 1+2M$ and let β_k be as in the proof of Lemma 2; indices with $B_k = 0$ contribute only $A_k\ln 1 = 0$. First, $a\ln(1+b/a) \le b$. Second, for $B_k > 0$ we have $\beta_k > 0$, and $\ln x \le x-1$ at $x = \beta_k^2(A_k+B_k)/(cB_k)$ gives

$$B_k\ln\Big(1+\frac{A_k}{B_k}\Big) \le B_k\ln c - 2B_k\ln\beta_k + \frac{\beta_k^2 A_k}{c},$$

since $\beta_k^2 \le 1 \le c$. Third, $\sum_k B_k(-\ln\beta_k) = \delta\sum_k(\beta_k-\beta_{k-1})(-\ln\beta_k) \le \delta\int_0^1(-\ln x)\,dx = \delta$, with $\beta_{-1} = 0$. Fourth, with $T_l = \sum_{k\ge l} A_k \le M\delta/\beta_l$ (terms with $\beta_l = 0$ vanish),

$$\sum_k A_k\beta_k^2 = \sum_l(\beta_l^2-\beta_{l-1}^2)T_l \le M\delta\sum_l\frac{(\beta_l-\beta_{l-1})(\beta_l+\beta_{l-1})}{\beta_l} \le 2M\delta.$$

Adding, $\sum_k\Phi(A_k,B_k)\ln 2 \le \delta + \delta\ln c + 2\delta + 2M\delta/c \le (4+\ln c)\delta$. In particular the constant in Theorem 4 may be replaced by $4/\ln 2 + \log_2(1+2\max(M,1))$.

For a set R of coordinates other than i, let x^R be x with the coordinates in R replaced by independent uniform bits, and put

$$W(R) = \mathbb{E}\,h(x)h(x^R) = \Pr[i\in U,\ U\cap R = \emptyset],$$
$$V(R) = \mathbb{E}g^2 - \mathbb{E}\,g(x)g(x^R) = \Pr[i\notin U,\ U\cap R \ne \emptyset].$$

$W(R)$ is the correlation of the pivotality of i, with its sign, at x and at x^R, and $V(R)$ is the influence of the block R on g. The two equalities are Parseval: $\mathbb{E}\,u(x)u(x^R) = \sum_{T\cap R=\emptyset}\hat{u}(T)^2$.

Theorem 4. *Let π be an order and $R_k = R_* \setminus S_k$ the set of coordinates not yet revealed after step k. Then $\sum_{l\le k} B_l = W(R_k)$ and $\sum_{l\ge k} A_l = V(R_{k-1})$. Consequently, if*

$$W(R_k)\,V(R_{k-1}) \le M\,\mathrm{Inf}_i(f)^2 \qquad \text{for } k = 1,\dots,m,$$

then $L_i^\pi(f) \le \big(1/\ln 2 + 4 + 2\log_2(1 + 2\max(M,1))\big)\mathrm{Inf}_i(f)$.

Proof. $\mathrm{lab}(T) \le k$ means $T \subseteq S_k$, that is $T \cap R_k = \emptyset$, which gives the first identity; $\mathrm{lab}(T) \ge k$ means $T \cap R_{k-1} \ne \emptyset$, which gives the second. For $K = 0$ the product in Lemma 2 is $(1-\delta)\hat f(\{i\})^2 \le \delta^2$, because $|\hat f(\{i\})| = |\mathbb{E}h| \le \Pr[h \ne 0] = \delta$. Apply Lemma 2 with $\max(M,1)$. □

In words: i is good along π if at every stage either the value of g is almost settled by the revealed coordinates (the resampled block has small influence on g), or the revealed coordinates say little about whether i is pivotal (resampling the rest destroys the pivotality of i). Because W decreases and V increases with R, a coarser chain $R_* = R'_0 \supset R'_1 \supset \cdots \supset R'_t = \emptyset$ that the order refines suffices: if $W(R'_{s+1})V(R'_s) \le M\mathrm{Inf}_i(f)^2$ for every s, the condition of Theorem 4 holds for every order that reveals $R'_s \setminus R'_{s+1}$ in block s.

Write $\mathcal{D} = \{h \ne 0\}$, $\mathcal{L} = \{g = -1\}$ and $\mathcal{H} = \{g = 1\}$. Since $g \in \{-1,0,1\}$ and $V(R) = \frac12\mathbb{E}(g(x) - g(x^R))^2$, every Boolean f has $V(R) = \Pr[x \in \mathcal{D},\, x^R \notin \mathcal{D}] + 4\Pr[x \in \mathcal{L},\, x^R \in \mathcal{H}]$; if f is monotone, then $h \ge 0$ and $W(R) = \Pr[x \in \mathcal{D},\, x^R \in \mathcal{D}]$. For monotone f the condition reads: the probability of staying in the pivotal band times the probability of jumping across it is $O(\Pr[\mathcal{D}]^2)$.

For example, let f be the OR of s disjoint ANDs of width w, let i lie in the first AND, let Q be the set of the other $w-1$ variables of that AND, and let $P_0 = (1-2^{-w})^{s-1}$ be the probability that no other AND is satisfied. Take the chain $R_* \supset Q \supset \emptyset$, that is, reveal the other ANDs first and Q last. Here $x \in \mathcal{D}$ when no other AND is satisfied and every variable of Q is true, so $\delta = \mathrm{Inf}_i(f) = P_0 2^{1-w}$ and $W(Q) = P_0 4^{1-w} = \delta^2/P_0$; with $V(R_*) \le 1$ the first product is at most δ^2/P_0. A move from $\mathcal{L}$ to $\mathcal{H}$ needs another AND to change, so resampling Q cannot make it, and $V(Q) \le \Pr[x \in \mathcal{D}] = \delta$; with $W(\emptyset) = \delta$ the second product is at most δ^2. The condition therefore holds with $M = 1/P_0$, which is bounded in the usual regime $s \approx 2^w \ln 2$, where P_0 is close to $\frac12$.

Conjecture 5 (filtration form)**.** *There is an absolute c such that every non-constant Boolean f has a coordinate i and an order π of the other coordinates with $L_i^\pi(f) \le c\,\mathrm{Inf}_i(f)$.*

By Theorem 1(a) and [9], Conjecture 5 implies the FEI conjecture with constant $c + 2/\ln 2$: the class of Boolean functions is closed under restriction, and the one-coordinate step applies at the coordinate given by the conjecture in every restriction. Conjecture 5 is a strengthening of (1); Section 7 shows no counterexample among the functions we can evaluate exactly, and Section 6 discusses where a single last element may not be enough.

4. Composition

Proposition 6. *Let B and Z be disjoint sets of coordinates, let $g\colon \{-1,1\}^B \to \{-1,1\}$ and $F\colon \{-1,1\} \times \{-1,1\}^Z \to \{-1,1\}$ be Boolean, and put $f(x_B, z) = F(g(x_B), z)$. Write $F(u,z) = F_0(z) + uF_1(z)$. For $i \in B$,*

$$\mathrm{Inf}_i(f) = \mathrm{Inf}_i(g)\,\|F_1\|_2^2,$$

$$E_i(f) = \|F_1\|_2^2\Big(E_i(g) - \Phi\big(\hat g(\emptyset)^2, \hat g(\{i\})^2\big)\Big) + \sum_{T\subseteq Z} \Phi\Big(\big(\hat F_0(T) + \hat g(\emptyset)\hat F_1(T)\big)^2,\ \hat g(\{i\})^2\hat F_1(T)^2\Big).$$

Proof. For $T_1 \subseteq B$ and $T_2 \subseteq Z$, $\hat f(T_1 \cup T_2) = [T_1 = \emptyset]\hat F_0(T_2) + \hat g(T_1)\hat F_1(T_2)$. For $i \notin T_1 \ne \emptyset$ the pair (p,q) at $T_1 \cup T_2$ is $\hat F_1(T_2)^2$ times the pair of g at T_1, so by homogeneity these terms sum

to $\|F_1\|^2$ times the terms of $E_i(g)$ with $T_1 \neq \emptyset$. The terms with $T_1 = \emptyset$ form the last sum. The influence is the sum of q. □

The last sum depends on g only through $\hat{g}(\emptyset)$ and $\hat{g}(\{i\})$. When g is itself a composition this can be iterated, which gives E_i and Inf_i of iterated block compositions from the spectra of the gates; the iteration for $g(h, \dots, h)$, carrying four numbers per leaf through each level, is in [10, Section 2], where it is used for towers of transitive gates. O'Donnell and Tan [7] prove a composition theorem for the conjecture itself.

5. A private AND

Proposition 7. *Let i and a set B of $k \geq 0$ other coordinates enter f only through $a = \mathbf{1}[x_i = x_b = 1$ for all $b \in B]$, so that $f = A(v) + c(v)\,a$ with v the remaining coordinates and $c \in \{0, \pm 2\}$. Let $\delta = \mathrm{Inf}_i(f) = 2^{-k} \Pr[c \neq 0] > 0$ and let π reveal v first and B last. Then*

$$2\big(1 - 2^{-k}\big) \leq \frac{E_i(f)}{\delta} \leq \frac{L_i^{\pi}(f)}{\delta} \leq 2\big(1 - 2^{-k}\big) + 2^{-k} \log_2 \frac{e\, 2^k}{\delta}.$$

Proof. In ± 1 notation $a = 2^{-(k+1)} \sum_{S \subseteq B \cup \{i\}} \chi_S$. Hence for $V \subseteq v$ and $S \subseteq B$, the coefficient of f at $V \cup S \cup \{i\}$ is $2^{-(k+1)} \hat{c}(V)$, and at $V \cup S$ it is the same number when $S \neq \emptyset$. Put $p_V = 4^{-(k+1)} \hat{c}(V)^2$; then $\delta = \sum_S q = 2^k \sum_V p_V$. Every $T = V \cup S$ with $S \neq \emptyset$ has $p(T) = q(T) = p_V$ and contributes $\Phi(p_V, p_V) = 2p_V$; there are $2^k - 1$ such S for each V, which gives $2(1 - 2^{-k})\delta$, and the remaining terms of E_i are nonnegative. In the witness, the label of $V \cup S$ with $S \neq \emptyset$ is the position of the last element of S; the r-th element of B is the last element of 2^{r-1} sets S, so this atom has $A = B = 2^{r-1} \sum_V p_V$ and contributes $2^r \sum_V p_V$, again $2(1 - 2^{-k})\delta$ in total. The atoms of the sets $T = V$ carry q-mass $Q = 2^{-k}\delta$ and p-mass $P = 1 - 2\delta + 2^{-k}\delta$, so $P + Q \leq 1$. By superadditivity their contribution is at most $\Phi(P, Q) \leq Q/\ln 2 + Q \log_2((P + Q)/Q) \leq Q \log_2(e/Q)$. □

Since $\delta = 2^{-k} \Pr[c \neq 0]$, the right-hand side is $2 + 2^{-k}\big(2k + \log_2 e - 2 + \log_2(1/\Pr[c \neq 0])\big)$, which is less than $2 + 2^{-k}\big(2k + \log_2(1/\Pr[c \neq 0])\big)$, so the ratio tends to 2 as the private AND grows, unless it is pivotal with probability below about 2^{-2^k}. The same computation applies to any private gadget whose Fourier coefficients at S and $S \cup \{i\}$ agree in absolute value for every non-empty S. Width-dependent bounds for every DNF, with no privacy assumption, are in [10].

6. Windows

For $1 \leq r \leq m$ let $\mathrm{lab}_r(T)$ be the set of the last r positions of π met by T (all of them if $|T| < r$), and

$$L_i^{\pi,r}(f) = H\big(\mathbf{1}[i \in U] \;\big|\; \mathrm{lab}_r(U \setminus \{i\})\big).$$

Then $L_i^{\pi,1} = L_i^{\pi}$, $L_i^{\pi,m} = E_i$, and

$$E_i(f) \leq L_i^{\pi,r}(f) \leq L_i^{\pi,r-1}(f),$$

because lab_{r-1} is a function of lab_r, which is a function of $U \setminus \{i\}$. The masses of the atoms of $L_i^{\pi,r}$ are signed sums, by inclusion and exclusion, of resampling correlations of derivatives of order at most r: for distinct coordinates $k_1, \dots, k_s$ other than i and a set R of coordinates disjoint from them, $\Pr[i \in U,\, k_1, \dots, k_s \in U,\, U \cap R = \emptyset] = \mathbb{E}\, D_{k_1} \cdots D_{k_s} h(x)\, D_{k_1} \cdots D_{k_s} h(x^R)$.

The ladder runs from Conjecture 5 at $r = 1$ to (1) at $r = m$. Window 1 sees only the last element of $U \setminus \{i\}$. When every variable lies in many overlapping certificates, the last element alone may carry little information about whether $i \in U$, while a pair of elements from one certificate of i may carry much more. We do not know whether Conjecture 5 holds, or whether it holds with some fixed window r in place of 1.

7. Exact values

The statements of Theorems 1 and 4, Lemma 2 and Propositions 6, 7 were checked with exact Walsh–Hadamard transforms. On 58 random non-constant functions of 3 to 7 variables (each value -1 with probability $\frac{1}{2}$, 0.2 or 0.05), at every coordinate and for every order when $n \le 6$ and 150 random orders when $n = 7$: the bounds of Theorem 1(a) hold to $3 \cdot 10^{-16}$, the shortest path of Theorem 1(c) agrees with the minimum over all orders to $3 \cdot 10^{-16}$, the martingale form and the resampling identities hold exactly, the bound of Lemma 2 holds with the smallest admissible M, and $E_i \le L_i^{\pi,3} \le L_i^{\pi,2} \le L_i^{\pi,1}$ with $L_i^{\pi,m} = E_i$. The composition formula on 60 random compositions to $4 \cdot 10^{-16}$, and the bracket of Proposition 7 on 80 random private ANDs. The table gives $F = \min_{i,\pi} L_i^\pi / \mathrm{Inf}_i$ and $\min_i E_i / \mathrm{Inf}_i$; $\mathrm{TRIBES}_{w\times s}$ is the OR of s disjoint ANDs of width w.

	AND_6	AND_8	AND_{10}	$\mathrm{TRIBES}_{2\times 3}$	$\mathrm{TRIBES}_{2\times 4}$	$\mathrm{TRIBES}_{2\times 5}$
F	2.2922	2.1048	2.0341	2.3160	3.0428	3.3892
$\min_i E_i/\mathrm{Inf}_i$	2.2922	2.1048	2.0341	2.3160	3.0428	3.3892
	MAJ_3	MAJ_5	MAJ_7	MAJ_9	MAJ_{11}	$\mathrm{TRIBES}_{3\times 3}$
F	1.0000	1.5204	1.7848	1.9578	2.0844	2.5474
$\min_i E_i/\mathrm{Inf}_i$	0	0	0	0	0	2.5474

On AND and tribes the witness of the best order equals E_i at the best coordinate. The functions AND_n are the case $v = \emptyset$ of Proposition 7, and they sit just below its upper bound: at $n = 10$, $F = 2.034057$ against 2.034068. On majority $E_i = 0$, because the spectrum lives on odd sets and $U \setminus \{i\}$ determines the parity of $|U|$, while the witness pays for the last element; its growth along n is slow and stays below 2.1 at $n = 11$.

The coordinate matters. For even k let $b(y) = (-1)^{N(y)}$ on $\{-1,1\}^k$, where $N(y)$ is the number of $j \le k/2$ with $y_j = y_{j+k/2} = -1$ (the inner product function), let $\mathbf{1} = (1,\dots,1)$, and let $f(y, x_i) = b(y)$ for $y \ne \mathbf{1}$ and $f(\mathbf{1}, x_i) = x_i$. For $k = 2, 4, 6, 8, 10$ the coordinate i has $E_i/\mathrm{Inf}_i =$ 2.672, 5.224, 7.387, 9.429, 11.439 and $\min_\pi L_i^\pi/\mathrm{Inf}_i =$ 2.950, 5.351, 7.426, 9.439, 11.442, while every coordinate j of y has $\min_\pi L_j^\pi/\mathrm{Inf}_j =$ 0.891, 1.699, 1.929, 1.983, 1.996. So E_i/Inf_i grows about linearly in k at i while every other coordinate stays below 2, and the choice of the coordinate in Conjecture 5 cannot be dropped.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [16] and Math Brain [14]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain and working alongside Codex agents running OpenAI's GPT-6.1 on the same problem, found the witness, its martingale form, the two-sequence condition and its resampling form, the composition formula, the private-AND bound and the windows. Claude Opus 5.5 (Anthropic) rederived each proof, wrote the numerical checks, prepared the manuscript and audited it. A further Claude agent found the bound of Remark 3 and checked it in Lean 4 with Mathlib (file in the code repository). These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-filtration-witness`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [14].

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